

Last time Normal subgroups and examples.

Definition $f : G \rightarrow G'$ homomorphism, define its **kernel** as

$$\ker(f) := f^{-1}(1_{G'}) = \{g \in G \mid f(g) = 1_{G'}\}.$$

Key observation $\ker(f) \triangleleft G$.

$\because 1_G \in \ker(f), x, y \in \ker(f) \implies f(xy) = f(x)f(y) = 1_{G'}, f(x^{-1}) = f(x)^{-1} = 1_{G'}^{-1} = 1_{G'}$.
Also $\forall g \in G, h \in \ker(f), f(ghg^{-1}) = f(g)f(h)f(g^{-1}) = 1 \implies g\ker(f)g^{-1} \subset \ker(f)$.

Examples

- Homomorphism $\text{sgn} : S_n \rightarrow \{\pm 1\}, A_n := \ker(\text{sgn}) \triangleleft S_n$. (alternating group)

$S_4 \triangleright A_4 = \{\text{id}, (2, 2), (3, 1)\}$ in terms of cycle lengths.

In fact $\sigma(a_1 \dots a_m)\sigma^{-1} = (\sigma(a_1) \dots \sigma(a_m))$, hence $\sigma A_4 \sigma^{-1} \triangleleft S_4$.

- $W : F$ -vector space, $\dim W < \infty$. $\det : \text{GL}(W) \rightarrow F^\times, \ker(\det) = \text{SL}(W) \triangleleft \text{GL}(W)$.

Observation $N := \ker(f : G \rightarrow G')$.

$\forall x, y \in G, f(x) = f(y) \iff f(xy^{-1}) = 1 \iff xy^{-1} \in N \iff x \in Ny \iff Nx = Ny$.

Hence f is injective $\iff \ker(f) = \{1\}$.

Next: Given $N \triangleleft G$, try to make G/N into a group s.t. the quotient map $q : G \rightarrow G/N, g \mapsto gN$ is a homomorphism.

Now $1_{G/N} = q(1_G) = N, \ker(q) = \{g \in G \mid gN = N\} = N$.

Definition-Proposition Let $N \triangleleft G$. On $G/N = \{\text{cosets } gN\}$, one can define a binary operation $(xN, yN) \mapsto (xy)N$, and:

- It makes G/N into a group,
- $q : G \rightarrow G/N$ is a homomorphism.

We say G/N is the **quotient group** / 商群 of G by N .

Proof Check xyN only depends on xN and yN .

Assume $xN = x'N, yN = y'N$. Then $x' = xu, y' = yv, u, v \in N$,

$$x'y'N = xuyvN = xy \underbrace{y^{-1}uyv}_{\in N} N = xyN.$$

Hence $(xN, yN) \mapsto xyN$ is well-defined.

Group axioms: reduce to G .

$$(xN, yN)zN = (xyN)zN = xyzN = xN(yzN) = (xN)(yN, zN).$$

Take $1_{G/N} := 1_G N$, then

$$xN 1_{G/N} = xN = 1_{G/N} xN.$$

Set $(xN)^{-1} = x^{-1}N$ since

$$xN x^{-1}N = 1_{G/N} = x^{-1}N xN.$$

To show q is a homomorphism, suffices to show

$$q(xy) = xyN = xN \cdot yN = q(x)q(y), \forall x, y \in G.$$

□

Remark The group structure on G/N is the unique one making q into a homomorphism.

$$\ker(q : G \rightarrow G/N) = N.$$

- Quotient of vector space V/U subspace = the quotient group $(V, +)/(U, +)$ together with multiplication by F ,
- Quotient ring R/I ideal = $(R, +)/(I, +)$ together with a multiplication.

Proposition (Universal property of quotient groups)

Let $f : G \rightarrow G'$ homomorphism, $N \triangleleft G$ s.t. $N \subset \ker(f)$. Then $\exists!$ homomorphism $\bar{f} : G/N \rightarrow G'$ s.t.

$$f = \bar{f} \circ q.$$

We say $\bar{f}$ is induced / 诱导 by f .

Proof Uniqueness: $\bar{f}q = f \iff \bar{f}(xN) = f(x), \forall x \in G \implies$ this determines $\bar{f}$.

Existence: If $xN = x'N, x, x' \in G$, then $x' = xu, u \in N \xrightarrow{N \subset \ker(f)} f(x') = f(x)f(u) = f(x)$.

Hence $\bar{f} : G/N \rightarrow G', xN \mapsto f(x)$ is a well-defined map.

$$\bar{f}(xN \cdot yN) = \bar{f}(xyN) = f(xy) = f(x)f(y) = \bar{f}(xN)\bar{f}(yN),$$

$\implies \bar{f}$ is a homomorphism. □

Corollary Let $f : G \rightarrow G'$ homomorphism, $N \triangleleft G, N' \triangleleft G', f(N) \subset N'$.

Then $\exists!$ homomorphism $\bar{f} : G/N \rightarrow G'/N'$ s.t. $q'f = \bar{f}q$. (see ref for commute graph)

Remark Take $N' = \{1\}, G' \simeq G'/\{1\}$ to get the previous Prop.

Proof Take $f' = q'f$ in previous Prop, $f(N) \subset N' = \ker(q') \implies N \in \ker(q'f)$. □

Corollary Let $f : G \rightarrow G'$ homomorphism. Recall $\text{im}(f) = f(G)$ is a subgroup of G' .

Put $N := \ker(f)$, we get $\bar{f} : G/\ker(f) \simeq \text{im}(f) \subset G'$.

Proof $\bar{f}(xN) = f(x)$ shows that $\text{im}(\bar{f}) = \text{im}(f)$.

If $\bar{f}(xN) = \bar{f}(yN)$ then $f(x) = f(y)$, so $xN = yN$. Hence $\bar{f}$ is injective $\implies \bar{f} : G/N \simeq \text{im}(f)$. □

In particular, surjective homomorphisms $\approx$ quotient groups.

Also $(G : \ker(f)) = |\text{im}(f)|$ by counting.

Examples

- $\text{sgn} : S_n \rightarrow \{\pm 1\}$ is surjective when $n > 1 \implies (S_n : A_n) = 2$.
- $N : \mathbb{H}^\times \rightarrow \mathbb{R}_{>0}^\times$ is a homomorphism, its kernel $\ker(N) := \mathbb{H}^1 \triangleleft \mathbb{H}^\times$.
 $R : \mathbb{H}^1 \rightarrow \text{SO}(3) = \{\text{rotations in } \mathbb{R}^3\}, q \mapsto R_q(x) = qxq^{-1}$ is a homomorphism, $\ker(R) = \{\pm 1\}$
Hence R induces $\bar{R} : \mathbb{H}^1/\{\pm 1\} \simeq \text{SO}(3)$.

Recall that $\mathbb{H} \simeq \left\{ \begin{pmatrix} z & -\bar{w} \\ w & \bar{z} \end{pmatrix} : z, w \in \mathbb{C} \right\}$ s.t. $N \leftrightarrow \det$.

Hence $\mathbb{H}^1 \simeq \left\{ \begin{pmatrix} z & -\bar{w} \\ w & \bar{z} \end{pmatrix} : |z|^2 + |w|^2 = 1 \right\}, \quad \{\pm 1\} \simeq \{\pm 1_{2 \times 2}\}.$

Set $\text{SU}(2) = \{\text{unitary } A \in M_{2 \times 2}(\mathbb{C}) \mid \det A = 1\}$, then

$$\text{SU}(2)/\{\pm 1\} \simeq \mathbb{H}^1/\{\pm 1\} \simeq \text{SO}(3).$$

Proposition Let $f : G \rightarrow G'$ surjective homomorphism. Then denote $N := \ker(f)$,

$$\{H' \subset G' \text{ subset}\} \longleftrightarrow \{N \subset H \subset G \text{ subset}\}$$

$$H' \mapsto f^{-1}(H'),$$

$$f(H) \longleftarrow H.$$

- $H'_1 \subset H'_2 \iff H_1 \subset H_2$,
- $H' \triangleleft G' (\iff H \triangleleft G)$, then $G \xrightarrow{f} G' \xrightarrow{\text{quot}} G'/H'$ induces $G/H \simeq G'/H'$.

Proof Clearly $H \subset f^{-1}(f(H))$. Also $f^{-1}f(H) \subset H$ since $f(x) = f(h), x \in G, h \in H \implies xh^{-1} \in N \implies x \in H, \therefore N \in H$.

In the other direction $H' = f^{-1}f(H)$ holds for any surjection $f : G \rightarrow G'$
 $\implies$ get $\simeq$, preservation of $\subset$ is clear.

Given $H, N \subset H \subset G, \forall g \in G, gHg^{-1} = H \iff \forall \bar{g} \in G', \bar{g}f(H)g^{-1} = f(H)$,
 $\therefore f$ surjective and $N \subset H$.

Finally, let $H \triangleleft G \iff H' \triangleleft G'$. Composite $G \xrightarrow{f} G' \twoheadrightarrow G'/H'$ is also surjective.
 Its kernel $= f^{-1}(H') = H \implies G/H \simeq G'/H'$. □

Example $\{\text{Subgroups of } \mathbb{Z}\} = \{m\mathbb{Z} \mid m \geq 0\}$.

$\forall n, \{\text{Subgroup of } n\mathbb{Z}\} \longleftrightarrow \{\text{Subgroups } n\mathbb{Z} \subset m\mathbb{Z} \subset \mathbb{Z}\} = \{m\mathbb{Z} : m \mid n\}$